Analytical DOS derivation for CNTs

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Abstract

In this tutorial, we derive analytically the density of states for a CNT tube that matches the expression in [1]. This tutorial is intended for students working with CNTs for the first time and fills a gap in the literature where the derivation of DOS is not explicitly shown.

1 Introduction

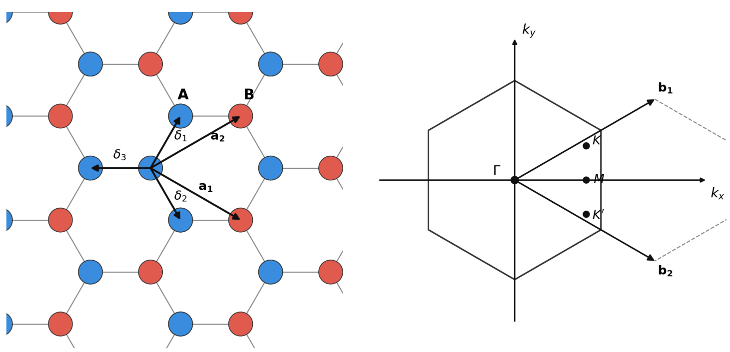

Figure 1: Graphene lattice and its Brillouin zone. [2]

Given the above coordinate choice for graphene unit cell, the band structure of a (N,0) CNT (Zig-Zag CNT), obtained from the tight-binding model of graphene [2], is expressed as (Obtained by putting the quantization condition $Nak_y = 2\pi m$ in the graphene band structure):

$$E = \pm t_0 \sqrt{f(k_x, m)} = \pm t_0 \sqrt{1 + 4\cos\left(\frac{3k_x a}{2}\right)\cos\left(\frac{\pi m}{N}\right) + 4\cos^2\left(\frac{\pi m}{N}\right)} \quad (1)$$

where $t_0 = 2.5eV$ is the tight-binding coupling constant, where a is the lattice constant and m is an index that runs from 0 to $N-1$. A thorough derivation can be found in [3], although with axes rotated.

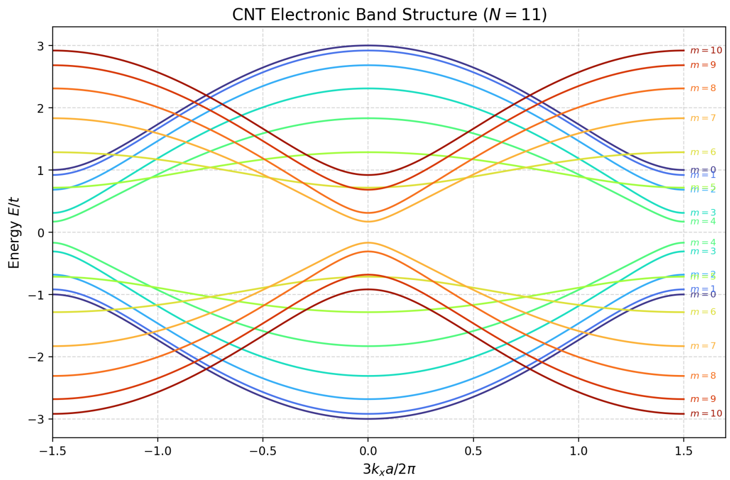

Figure 2: Electronic band structure of an $N = 11$ zig-zag CNT computed from (1), with each curve labelled by its sub-band index $m = 0, 1, \ldots, N-1$.

1.1 Analytical expression for band-gap

CNT is one of those nice systems where the band-gap can be expressed analytically. In (1), consider the bands with a $+$ sign. Then the band-minimum, E_{min_m} for a given band index, m, occurs when

$$\partial_{k_x} f(k_x, m) = 0 \implies \frac{3k_x a}{2} \in \{0, \pi\} \tag{2}$$

while the minimum of the set E_{min_m} across all bands, as m is varied is obtained when the quadratic equation in $\cos(\frac{\pi m}{N})$ is minimized:

$$\begin{aligned} \partial_{\cos(\pi m/N)} f(k_x, m)\Big|_{\frac{3k_x a}{2} \in \{0,\pi\}} &= 0 \\ \implies \cos\left(\frac{\pi m}{N}\right) &= -\frac{1}{2}\cos\left(\frac{3k_x a}{2}\right)\Big|_{\frac{3k_x a}{2} \in \{0,\pi\}} \\ &= \pm\frac{1}{2} \end{aligned} \tag{3}$$

which means $\cos\left(\frac{\pi m}{N}\right)$ needs to be as close to $\pm\frac{1}{2}$ as possible. This is exactly possible whenever $N = 3m$, then band-gap is exactly 0 and we call such a CNT metallic. Though in this article, we will consider only CNTs with a bandgap so let us restrict ourselves to when $N = 3m - 1$, i.e $N = 8, 11, 14, 17$ etc. A representative band-diagram is shown for $N = 11$ below:

Given a value of N, m will take values in the set $\{0, 1, 2, \ldots, N-1\}$. Let

$$m_0 := \arg\min_m \left\{ \left| m - \frac{N}{3} \right|, \left| m - \frac{2N}{3} \right| \right\}.$$

The band-minima is then written as (pairing the value of m_0 with $\frac{3k_x a}{2} = \pi/0$):

$$E_g = 2t_0 \sqrt{1 - 4\cos\left(\frac{\pi m_0}{N}\right) + 4\cos^2\left(\frac{\pi m_0}{N}\right)}. \tag{4}$$

1.2 Analytial expression for Density of States

Good point to obtain an expression for the density of states near the band-minimum. First, we need to show the following approximation holds near the band-minimum:

$$\frac{d}{dE}\left(\frac{3k_x a}{2}\right) \approx \frac{|E|}{t_0\sqrt{E^2 - (E_g/2)^2}} \tag{5}$$

In section 1.1 we discovered that a neighbourhood of band minimum comprises of m_0 with $\frac{3k_x a}{2} \approx \pi$. So equation (1) around band-minimum can be rewritten as a taylor expansion around $\frac{3k_x a}{2} = \pi$: (Hereafter, k_x refers to the small-signal expansion around $k_x = \frac{2\pi}{3a}$)

$$E \approx \pm t_0 \sqrt{1 + 4\cos\left(\frac{\pi m_0}{N}\right)\left(-1 - \frac{1}{2}\left(\frac{3k_x a}{2}\right)^2\right) + 4\cos^2\left(\frac{\pi m_0}{N}\right)} \tag{6}$$

Plugging in the value of E_g from (4), we get:

$$E \approx \pm t_0 \sqrt{\left(\frac{E_g}{2t_0}\right)^2 - \left(2\cos\left(\frac{\pi m_0}{N}\right)\right)\left(\frac{3k_x a}{2}\right)^2} \tag{7}$$

Since m_0 takes $\cos(\frac{\pi m_0}{N})$ to be as close to $\frac{1}{2}$ as possible, for a big enough N, we can approximate the above equation as:

$$E \approx \pm t_0 \sqrt{\left(\frac{E_g}{2t_0}\right)^2 - \left(\frac{3k_x a}{2}\right)^2} \tag{8}$$

Rerrangment gives:

$$\frac{3k_x a}{2} \approx \pm \frac{1}{t_0}\sqrt{E^2 - \left(\frac{E_g}{2}\right)^2} \tag{9}$$

(5) follows from differentiating (9) with respect to E.

From here, the calculation of density of states is straightforward. Density of States is defined as:

$$D(E) = \frac{1}{L} \frac{\text{\# States in [E, E+ dE]}}{dE} \tag{10}$$

For a periodic system, this means counting the number of values of k_x whose energy lies in the range $[E, E + dE]$ i.e. $dk(E)$. We already know $dk(E)$ from (5).

k-states are spaced uniformly with space

$$d\left(\frac{3k_x a}{2}\right) = \frac{3\pi a}{L} \tag{11}$$

Now we compute $D(E)$ in steps, first we find the numerator ie # states in [k, k+dk]:

$$dk(E) = \frac{d}{dE}\left(\frac{3k_x a}{2} dE\right) \approx \frac{|E|dE}{t_0\sqrt{E^2 - (E_g/2)^2}} \tag{12}$$

The number of states in this range of dk is given by dividing above equation by 11

$$\frac{1}{\frac{3\pi a}{L}} \frac{d}{dE}\left(\frac{3k_x a}{2} dE\right) \approx \frac{1}{\frac{3\pi a}{L}} \frac{|E|dE}{t_0\sqrt{E^2 - (E_g/2)^2}} \tag{13}$$

Dividing by L gives:

$$D(E) \approx \frac{1}{3\pi a} \frac{|E|}{t_0\sqrt{E^2 - (E_g/2)^2}} \tag{14}$$

1.2.1 Degeneracy

But the equation above is not yet correct, since there is an eight-fold $(2 \times 2 \times 2)$ degeneracy in a CNT:

1. In (1), (k_x, m) and $(-k_x, N - m)$ give the same energy (a factor of 2).
2. Electron spin takes two values (a factor of 2).
3. Each band is symmetric about its minimum, and our expression considered only a single branch of the expansion around the band-minimum (a factor of 2).

The last degeneracy can also be seen in the figure 2 in Introduction section. The final Density of States is then given as:

$$D(E) \approx \frac{8}{3\pi a} \frac{|E|}{t_0\sqrt{E^2 - (E_g/2)^2}} \tag{15}$$

1.2.2 Verification

Let us derive the surface charge density expression Each state has its charge distributed uniformly across the cylinder. So that the surface charge density is given as (remember DoS is already per unit length, so we just need to divide by the circumference of the cylinder while computing surface charge density):

$$\sigma = \frac{q}{2\pi R} \int_{E_1}^{E_2} D(E) f(E) dE \tag{16}$$

$$\sigma = \int_{E_1}^{E_2} \frac{8q}{6\pi^2 Ra} \frac{|E|}{t_0 \sqrt{E^2 - (E_g/2)^2}} f(E) dE \tag{17}$$

From geometry, we know the relation:

$$2R \sin \frac{\pi}{N} = \sqrt{3} a \tag{18}$$

For big enough N, we can linearize sin function and this gives:

$$\sigma = \int_{E_1}^{E_2} \frac{8q}{3\sqrt{3}\pi a^2 N} \frac{|E|}{t_0 \sqrt{E^2 - (E_g/2)^2}} f(E) dE \tag{19}$$

This agrees with the expression we have in Eq. 15 of [1]

References

[1] Francois Leonard and Derek A. Stewart. Properties of short channel ballistic carbon nanotube transistors with ohmic contacts. *Nanotechnology*, 17(18):4699–4705, 2006. arXiv:cond-mat/0608616.

[2] A. H. Castro Neto, F. Guinea, N. M. R. Peres, K. S. Novoselov, and A. K. Geim. The electronic properties of graphene. *Reviews of Modern Physics*, 81(1):109–162, 2009. arXiv:0709.1163.

[3] Christian Schönenberger. Bandstructure of Graphene and Carbon Nanotubes: An Exercise in Condensed Matter Physics. Lecture notes, Nanoelectronics Group, University of Basel, 2015. Corrected 2015; originally April 2000. `https://nanoelectronics.unibas.ch/wordpress/wp-content/uploads/teaching/LCAO-band-structure-graphene-CNTs.pdf`.